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PI Name:	Jacobs, Jeffrey Ph.D.		
Project Title:	An Experimental Investigation of Incompressible Richtmyer-Meshkov Instability		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jwjacobs@u.arizona.edu	**Fax:**	FY 520-621-8191
PI Organization Type:	UNIVERSITY	**Phone:**	520-621-8459
Organization Name:	University of Arizona		
PI Address 1:	PO box: 210119		
PI Address 2:	Aerospace & Mechanical Engineering		
PI Web Page:			
City:	Tucson	**State:**	AZ
Zip Code:	85721	**Congressional District:**	7
Comments:			
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Contact Monitor:	Niederhaus, Charles	**Contact Phone:**	(216) 433-5461
Contact Email:	charles.niederhaus@grc.nasa.gov		
Flight Program:			
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Richtmyer-Meshkov instability occurs when a planar interface separating two fluids of different densities is impulsively accelerated in the direction of its normal. The focus of this research project is to observe and quantify the nonlinear development of this instability from a simple well characterized initial state into a turbulent flow, and to determine whether a self-preserving final turbulent state is achieved. The experiments utilize a novel experimental apparatus which circumvents many of the experimental difficulties that have previously limited the study of Richtmyer-Meshkov instability. In this system, the instability is generated by impulsively accelerating a rectangular box containing two liquids. The effects of gravity are minimized by keeping the fluids in free-fall. Previous NASA funding supported the construction of a laboratory apparatus in which the impulsive acceleration is imparted by bouncing the fluid box off of a fixed spring. The current project will extend the previous study and focus on the turbulent regime to determine self-similarity and to obtain the turbulent growth rate. The extended duration of microgravity offered by the 2.2 Second Drop Tower at GRC and the KC-135 Reduced Gravity Aircraft will be utilized to study the turbulent regime. An

Task Description:	adaptation of the existing laboratory apparatus for use in the 2.2 Second Drop Tower was initiated under previous NASA funding. Experiments will therefore initially be carried out in this facility using the new apparatus and later progress to the Reduced Gravity Aircraft. The adaptation of the experiments to the KC-135 will be aided by experiments at the University of Arizona that will focus on developing techniques for initial interface formation using paramagnetic fluids. These methods will also prove useful for a possible future space experiment. The objective of the funded research is to carry out ground-based experiments to study the nonlinear development and transition to turbulence of Richtmyer-Meshkov instability with the goal of quantifying the development of the resulting turbulent flow and to determine if a self-similar state is achieved. The long term objective of this project is to develop methods and procedures for a possible future flight experiment in order utilize the low gravity environment of space for the creation of an experiment with virtually unlimited duration to study the fully turbulent Richtmyer-Meshkov instability. Richtmyer-Meshkov instability is one of the most fundamental of fluid instabilities and is of importance in fields ranging from astrophysics to materials processing. For example, Richtmyer-Meshkov instability is of critical importance to inertial confinement fusion. The degree of compression capable in laser fusion experiments is ultimately limited by Richtmyer-Meshkov instability and another similar instability called Rayleigh-Taylor instability. Thus, these instabilities represent the single most significant barrier to attaining positive-net-yield fusion reactions in laser fusion facilities. Richtmyer-Meshkov instability is also of importance to high speed combustion applications, where, for example, it can greatly enhance mixing and thus significantly alter the burning rate. Knowledge gained from this study should aid in the design of laser fusion targets and supersonic combustors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Work the past year has been focused in three main areas: 1) The continuation of experiments in the GRC 2.2 Second Drop Tower, 2) The continuation of experiments in the University of Arizona 3 Meter Drop Tower, and 3) The development of methods for producing stable interfaces in microgravity using magnetic fluids. Experiments were conducted for the second year in NASA Glenn's 2.2 Second Drop Tower. Previous experiments have shown that the time duration and vibration incurred in closing the drag shield and raising the rig to the top of the tower would allow the interface between the two fluid to mix. In addition, the interface conditions were not repeatable. As a result, modifications were made to the rig to allow for withdrawal of the thin mixed region at the interface just prior to the rig drop. Containers of the two fluids are mounted in the rig, and the two fluids are gravity fed into the tank at the top and bottom separately. The excess fluid is withdrawn through small holes at the four corners of the tank at the interface location into a waste container, thus removing any mixed fluid. This modification allowed for a sharper, more repeatable interface. Experiments were conducted using this configuration over a range of Reynolds number lower than possible with the Arizona drop tower. Initial comparison of nondimensional measured amplitude with earlier data from the University of Arizona drop tower show good agreement. High Reynolds number experiments were difficult to conduct due to safety considerations in the drop tower, and unexpected difficulties in obtaining a well-defined initial perturbation when oscillating the tank at higher amplitudes. The difficulties arose because the drag shield is held by a single cable, and internal oscillations would induce secondary oscillations of the drag shield, including undesirable out-of-plane motions. These difficulties also made it difficult to conduct experiments with shorter wavelengths. The next planned phase is to work with a 3D perturbation, where the tower drawbacks are less of a concern, and the extended time duration could prove even more beneficial. Work is continuing on the University of Arizona apparatus built under funding by the previous grant. These new experiments focus on three-dimensional Richtmyer-Meshkov instability. A new square tank has been constructed which is mounted to the apparatus oriented at a 45 degree angle with respect to the alignment of the original rectangular tank. In this configuration the standing wave forcing mechanism translates the tank in a direction of the new tank's diagonal which is effective for producing square-type 3D perturbations. Experiments using this configuration have been performed focusing on perturbations with 1 1/2 and 2 1/2 wavelengths within the tank. The new experiments are being conducted using PLIF with the light sheet oriented along the tank diagonal. In addition a small number of experiments have been performed in which the entire interface is visualized by making one of the fluids opaque (by adding titanium dioxide particles) and using flood illumination. Recent experimental results show the three-dimensional instability to grow much faster in the nonlinear regime than was previously found with two-dimensional experiments, a result in agreement with published models. However, it is notable that these models show both 2D and 3D patterns to have the same late-time dependence of the growth rate decaying as 1/t as t?8. In contrast, our new experiments indicate constant growth rate at late times. The constant growth rate can be attributed to basic differences in the vorticity distributions of the two cases. The two dimensional instability results in an array of vortices of alternating sign. Thus the growth in amplitude can be attributed to the motion induced by this stationary vortex array. The three-dimensional instability results in arrays of vortex rings, alternately directed upward and then downward. Thus in this configuration the vorticity distribution is not stationary; The vortex rings themselves move upward and downward yielding faster growth. This, finding is important because it contradicts the accepted model for the late-time behavior of this instability. Work is also continuing on the development of our capability to produce stable interfaces in microgravity using magnetic fluids. Earlier work in this effort has concentrated on evaluating electromagnet configurations for this purpose, modeling them using the finite element package ANSYS. Using the results of this analysis it was decided to implement a dipole electromagnet configuration and we have recently installed a large (10" pole diameter) electromagnet with power supply. More recent work has been focused on using ANSYS to determine the optimum pole face geometry to carry out the dipole experiments. In this study field distributions were computed to determine the best pole contour for providing a large field gradient while also providing nearly horizontal field contours. The results of this research have indicated that a cosine pole face satisfies our requirements and have recently completed the machining of this geometry onto the poles of our magnet. Preliminary experiments have begun in which we have been able to stabilize a heaver magnetic fluid on top of a lighter magnetic one. In addition, it appears that our calculations have correctly predicted that our choice of pole geometry does generate a flat interface.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
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Presentation	Chapman, P.; Jacobs, J. W. "Experiments on the Three-Dimensional Incompressible Richtmyer-Meshkov Instability" 56th Meeting of the APS/DFD, East Rutherford, New Jersey Nov-2003
Presentation	Jacobs, J. W.; Chapman, P. R; Wilkinson, J. P. "Experiments on the Three-Dimensional Richtmyer-Meshkov and Rayleigh-Taylor Instabilities" 9th International Workshop on the Physics of Compressible Turbulent Mixing, Cambridge, UK Jul-2004